Internship Report

On

Signal & Telecommunication Department

At

Divisional Railway Manager’s Office, New Delhi

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Submitted by

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Acknowledgement

I would like to offer my sincerest thanks to many individuals who have aided and assisted me during my summer internship. Without their assistance, this experience would not have been the same.

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I would also like to extend my gratitude to all the staff of the DRM Office, New Delhi. Their hospitality, support, and teamwork made me feel like an integral part of the team right from the very beginning. I appreciate the chance to have collaborated with and learned from such a brilliant set of professionals.

My heartfelt gratitude also to the **Signal & Telecommunication department of the DRM Office, New Delhi**, for giving me this great internship opportunity. The experience has been extremely enriching, and I have learned hands-on skills and information on the railway signalling and telecommunication industry that I will take with me for the rest of my career.

Lastly, I would like to thank friends and family for their endless support and encouragement throughout the process.

Abstract

This is the account of work and learning experience in a summer internship carried out under the **Signal & Telecommunication Department, DRM Office, New Delhi**. The primary task of this internship was to acquire practical experience and on-job knowledge in the important area of railway telecommunication and signalling, systems which play a key role in the safe and efficient running of the Indian Railways.

The internship was about familiarization with the concepts of **contemporary signalling systems, maintenance processes of telecommunication hardware, UTS server and Kavach project**. The important activities were the study of **Route Relay Interlocking (RRI) functioning, watching the maintenance of Automatic**

Signalling systems, helping in testing lines of communications and field visit to observe the Kavach project personally. A major portion of the training was spent on the project, **"Kavach"** which includes the **automated braking systems equipped in the new Locomotives through the use of RFID**.

During the internship, my understanding of **electronic interlocking, optical fiber communication, and train detection mechanisms** has improved significantly. The report captures the technical know-how obtained, jobs done, and practical skills acquired. This experience has given insight into the application of engineering principles in the railway industry and has reaffirmed my passion in this area.

Introduction

Overview of Indian Railways

Indian Railways (IR) is among the world's largest and busiest rail network, being a vital pillar of the nation's infrastructure as well as economy. Formed during the British colonial period, it has evolved into a gigantic organization connecting the length and breadth of the country, carrying millions of passengers and tons of goods on a daily basis. Being the lifeblood of India, IR is critical to social and economic growth, promoting trade, tourism, and interstate travel for a multiregional population. The system is structured into various zones and divisions for proper management, with the DRM Office in New Delhi belonging to the Northern Railway zone, which is one of the most visible zones in the nation. Its aspiration for modernization and growth continues to propel India's growth narrative.

Role of the Signal & Telecommunication Department

Signal & Telecommunication (S&T) Department is the brain of railway operations, and it is responsible for the vital duty of ensuring safe, efficient, and timely movement of trains. The main task of the department is to plan, fit, and ensure the sophisticated systems controlling train movements and communication within the network. This ranges from the signals that motorists observe, to interlocking systems that obviate conflicting movement at intersections, and telecommunication networks that link stations, control offices, and administrative buildings. The S&T Department leads the technology adoption efforts in Indian Railways, with a consistent effort to advance infrastructure to improve safety standards, raise line capacity, and enhance overall operational performance.

Objectives of the Internship

The main aim of this internship was to establish a correlation between theoretical knowledge obtained in academia and actual engineering practices in the specialized area of railway operations. The major objectives were:

- To acquire practical, on-ground experience with the different signalling and telecommunication systems employed in Indian Railways.
- To comprehend the day-to-day working, maintenance processes, and safety measures adopted by the S&T Department.
- To learn about the architecture and operation of new railway technologies such as Electronic Interlocking and Optical Fiber Communication.
- To apply engineering concepts to a project and make a useful contribution to the work of the department.
- To acquire professional skills like problem-solving, collaboration, and technical documentation within a large public sector organization.

Scope and Structure of the Report

This internship report covers the experience and knowledge acquired during the summer internship at the Signal & Telecommunication Department, DRM Office, New Delhi. The report gives a detailed description of the technical systems and operational processes seen.

The report is divided into five chapters. Chapter 1 is an introduction to Indian Railways and the S&T Department, as well as the aim of the internship. Chapter 2 and 3 focus on the technical basics of railway signalling and telecommunication systems, respectively. Chapter 4 explains the particular on-site activities, duties discharged, and the work undertaken on the project assigned. Chapter 5, lastly, concludes the report with an abstract of major learnings, skills learned, and insights on the overall experience.

Fundamentals of Railway Signalling

Basic Principles and Importance of Signalling

Railway signalling refers to a system of guiding railway traffic and preventing trains from being in close proximity to one another at all times. The underlying principle of signalling is to have a safe distance between trains, also referred to as the "space interval system". This helps to prevent collisions as well as facilitate the smooth and efficient flow of rail traffic. Signalling systems are crucial to safety, avoiding accidents by giving drivers precise directions on the condition of the track in front of them. In addition, a good signalling system has a direct effect on line capacity; reducing the space between trains safely allows more trains to travel on a piece of track, enhancing the efficiency and punctuality of the network as a whole.

Types of Signals and Their Aspects

Signals are the main communication between the control system and the train driver. They give information on the state of the track ahead. A signal's "aspect" is its visible appearance (e.g., the colour of a light or arm position), and the "indication" is the instruction given.

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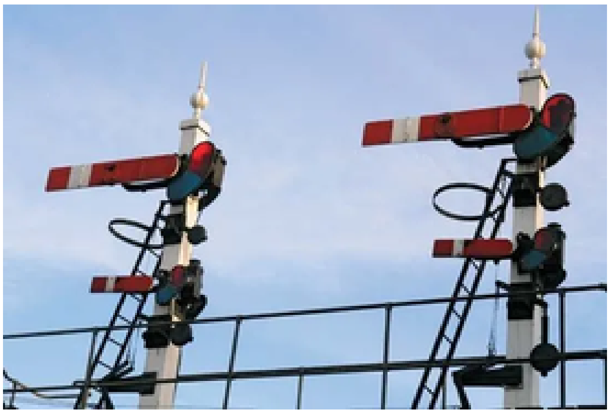

- **Semaphore Signals:** These are traditional mechanical signals made up of a movable arm. The location of the arm (horizontal, slanted, or vertical) shows whether the driver has to stop, go with care, or go at normal speed. They are typically employed in areas where there is not much traffic.

Fig.1 - Semaphore Signals

- **Colour Light Signals:** They are the most widely used form of signals in contemporary railways. They employ coloured lights (red, yellow, and green) to indicate instructions.

- **Red (Stop):** Tells a driver to stop the train before reaching the signal.
- **Yellow (Caution):** Instructs a driver to slow down and be ready to stop at the following signal.
- **Green (Proceed):** Shows that the way ahead is clear and the driver may proceed at the allowed maximum speed.
- **Double Yellow:** Frequently employed on high-speed roads to give a warning to decrease speed earlier.

Introduction to Interlocking Systems

Interlocking is a key safety system that maintains the integrity of a route set for a train. It is a signal apparatus system that prevents conflicting movements through an arrangement of tracks like junctions or crossings. The system is meant to prevent the clearance of signals for a train unless the route has been shown to be safe. For instance, an interlocking system will not allow a signal to display 'proceed' if a group of points (switches) on the track is not properly aligned or if another train is on a conflicting path. Early systems were mechanical, but contemporary systems are comprised of:

- **Relay Interlocking (RI):** Uses electrical relays and hard-wired logic circuits.

Fig.2 – Relay Interlocking

- **Electronic Interlocking (EI):** A microprocessor or computer-based system that employs software logic, providing greater flexibility, reliability, and diagnostic features.

Fig.3 – Electronic Interlocking

Train Detection Mechanisms

To safely control traffic, the system has to be aware where trains are. This is done by train detection mechanisms.

- **Track Circuits:** One part of track is electrically insulated, and a small amount of voltage current is flowed through the rails. As wheels and axle of a train run through this part, they cause a short circuit, diverting the current. This reduction in current is sensed, showing that the section of track is occupied.
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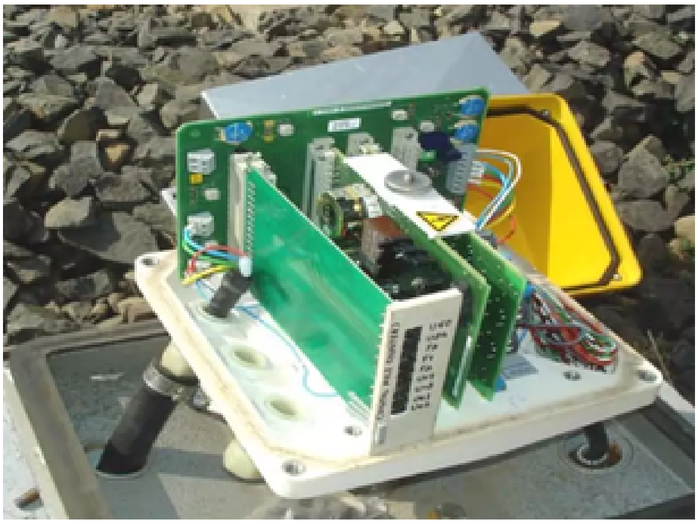

- **Axle Counters:** They are placed at the entrance and exit of a track section. They measure the axles entering the section and compare them with the axles exiting. The section is deemed to be clear if the count is the same. Axle counters are more effective in places where track conditions are poor or AC traction is employed.

Fig.4 – Axle Counter

Modern Signalling Systems

The Indian Railways keeps enhancing the signalling infrastructure to reduce risk and increase capacity.

Automatic Signalling: Under this scheme, the whole track is segmented into a sequence of automatic block sections. The signals controlling these sections are automatically controlled by the movement of trains (using track circuits or axle counters). The provision enables trains to be spaced more closely and is required for high-density suburban and mainline routes.

Centralized Traffic Control (CTC): It is a system in which the signal and point control for an extensive railway area is consolidated into one control room. It offers a bird's eye view of the whole section and facilitates effective and adaptable management of train traffic.

Railway Telecommunication Systems

Importance of Communication in Railway Operations

While signalling systems ensure the safe separation of trains, telecommunication systems provide the essential voice and data connectivity that underpins all railway operations. Effective communication is vital for coordinating train movements, managing emergencies, and ensuring administrative efficiency. A robust telecom network allows for real-time information exchange between train drivers, station masters, and central control rooms, making it an indispensable component of a modern, safe, and reliable railway. It supports everything from operational voice calls to data transmission for signalling and passenger information systems.

Overview of Communication Media

The foundation of the railway telecommunication network is established through different transmission media with a high regard for reliability and bandwidth.

- **Optical Fiber Cable (OFC):** It is the main mode of communication employed in Indian Railways. OFC provides very high bandwidth, resistance to electromagnetic interference (from power lines over the railway track), and security. OFC is laid parallel to the tracks and becomes a high-speed digital network that conveys data for signalling, voice communication, and other administrative applications.
- **Quad Cable:** It is comprised of pairs of copper conductors twisted together. Although being substituted by OFC in most fields, it continues to find application for last-mile connectivity and for most local circuits, such as communication to signals, trackside equipment, and level crossing gates. It is a trustworthy medium for voice-frequency communication.

Control and Administrative Communication Networks

Railway communications can be broadly classified into networks for train operations (Control) and those for general administration.

- **Control Communication:** This is a mission-critical network employed solely for the purpose of controlling train traffic. It links the central controller with station masters, drivers, and guards between two specific stations along a particular route. The most prevalent system used is the omnibus circuit, in which all stations along a section are able to communicate with the central controller. This provides immediate and effective coordination for train movement.
- **Administrative Communication:** The network sustains the on-going administrative activities of the railways. It comprises telephone exchanges (such as a railway staff private telephone network), data networks for computer systems (for ticketing, freight management, etc.), and video conferencing facilities.

Data Communication and Passenger Information Systems

Contemporary railway operations are very much data communication-dependent for operational as well as passenger-oriented services.

- **Data Loggers:** These are units placed at signalling centers to log and track all signalling events in real time (e.g., signal changes, track circuit operation, point movements). This information is very important for fault analysis, maintenance, and accident investigation.

- **Passenger Information Systems (PIS):** This system offers real-time train information to passengers. It covers:
- **Coach Guidance Displays:** Electronic boards on platforms showing the location of coaches for an arriving train.
- **At-a-Glance Displays:** Station concourse large screens displaying arrivals and departures.
- **Automated Announcements:** Computer-generated or pre-recorded voice announcements relating to train status.

They are supplied with data from the central control and signalling network, providing accurate and timely information to passengers.

Internship Experience and Project Work

Departmental Structure and Workflow

The Signal and Telecommunication (S&T) Department in the Divisional Railway Manager (DRM) Office, New Delhi, serves as the nerve centre for the train control and communication system of the region. The department's organization is carefully planned for safety as well as efficiency. At the top sits the Senior Divisional Signal and Telecom Engineer (Sr. DSTE), who is in charge of strategic planning and project management. Assisting them are Divisional Engineers (DSTE/ADSTE), each with designated areas or projects under their charge.

The operational process is a repetitive cycle of planning, execution, and reporting. Instructions for large-scale upgrades, like commissioning an Electronic Interlocking system or introducing KAVACH on a large scale, are received from the Zonal Headquarters. The office of the Sr. DSTE converts these instructions into executable plans, which are then cascaded to the field units. These units, headed by Senior Section Engineers (SSE) and assisted by Junior Engineers (JE), constitute the frontline of the department. They oversee teams of technicians dealing in fields such as Relay Interlocking, Electronic Interlocking, Axle Counters, and OFC maintenance. The day starts with a "tool-box" meeting where work is delegated and safety procedures are discussed. Any failure of equipment is recorded in a Signal Failure Memo, which initiates a systematic troubleshooting process, with escalations controlled through this hierarchical chain to ensure quick resolution.

On-Site Observations and Practical Training

Much of the internship comprised on-site visits and hands-on training under guidance by seasoned engineers and technicians. These sessions were invaluable in bridging the divide between theoretical understanding and actual practice. My onsite observations included:

- **Relay and EI Rooms Visit:** I spent a lot of time in Relay Interlocking (RI) and Electronic Interlocking (EI) rooms. In the RI rooms, I followed the hardwired logic traces through arrays of electromagnetic relays, enjoying how tangible connections between relay contacts implement fail-safe logic (for instance, demonstrating that a point is set and latched prior to the control relay for the corresponding signal being energized). The EI rooms illustrated modern systems' transition. In these, the logic is software, executing on redundant, hot-standby industrial computers. I was taught to read the event log on the Visual Display Unit (VDU), which shows a real-time schematic of the whole yard and is used as the main diagnostic tool by the engineering

- team.

Fig.5 – Relay Room

- **Field Equipment Maintenance:** I joined maintenance crews on their routine round-ups. This included being taught how to use a multimeter to test the operating current and detection voltage of an IRS-type point machine. I also shared in the upkeep of track circuits, knowing the basic concept of the train axle shunting the low-voltage current between rails. We maintained both DC track circuits used in station yards and newer Audio Frequency Track Circuits (AFTC) on the main line, which provide longer block sections and improved immunity to interference.
- **Telecommunication Network Test:** I was trained by observing the OFC maintenance crew. I saw the mechanism of fusion splicing, which involves two optical fibres being accurately cleaved and then melted end to end using an electric arc to form a nearly lossless join. I discovered the crucial role of Optical Time-

Domain Reflectometer (OTDR) testing in locating the exact position of a break within a cable, sometimes to the sub-meter level, which is key to fast network restoration.

Detailed Project Report: Kavach

My core project was an in-depth study and analysis of the KAVACH Automatic Train Protection system.

Project Introduction and Objectives

KAVACH is India's response to the international need for secure rail operations. It is a homegrown ATP system created by the Research Design and Standards Organisation (RDSO). The main goal of the project was to know KAVACH as not merely a safety device, but as a full-fledged train control system.

Its fundamental safety requirement is to meet Safety Integrity Level 4 (SIL-4) standard, which demands that the likelihood of an unsafe failure be exceedingly low (in the range 10^{-8} to 10^{-9} per hour). This is accomplished by a fail-safe design philosophy and redundancy to a high degree. The aims of the system extend beyond preventing accidents; it is intended to increase line capacity by allowing trains to be operated more closely together in safety and to increase punctuality by offering loco pilots continuous cab signalling, essential for operating at high speeds, particularly in poor visibility such as fog where conventional lineside signals cannot be seen.

Methodology and Tasks Performed

My approach was systematic, starting with foundational theory and progressing to hands-on analysis of the live system.

- **System Architecture Study:** I conducted a component-level study of the KAVACH ecosystem:
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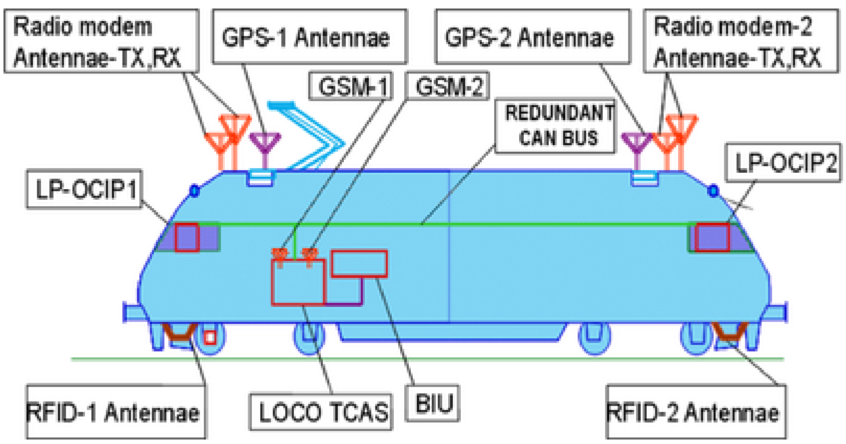

- **Onboard Unit (Loco KAVACH):** It is the brain of the locomotive. It contains a critical computer which permanently runs the central safety logic. Its inputs are information from a radio modem, GNSS receiver for position, tachometers for accurate speed measurement, and an RFID reader. Its outputs send commands to the Driver Machine Interface (DMI) and, most importantly, the Brake Interface Unit.

Fig.6 – Onboard Unit

- **Trackside Units:** These are the ground infrastructure. Passive RFID tags, preprogrammed with individual location and track geometry information (gradients, curves), are attached to the track sleepers. The Station KAVACH unit is an essential computer that communicates directly with the station's interlocking system (either by reading relay contacts in an RI system or through

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- a data link in an EI system) to obtain real-time signal aspect information.

Fig.7 – Trackside Unit

- **Communication Network:** KAVACH utilizes a strong UHF radio network that works in the 400 MHz band. It employs a Time Division Multiple Access (TDMA) protocol, with a single base station being able to handle communication with multiple trains within its range effectively, allowing data exchange to happen in a timely and interference-free manner.
- **Observation of Installation and Commissioning:** I witnessed the complete life cycle, from physical installation of trackside balises (RFID tags) at pre-computed braking distances from signals, to the involved process of commissioning. Commissioning entails extensive end-to-end testing and preparation of a "Safety Case," a formal report that includes evidence and arguments to demonstrate that the system is sufficiently safe for operational use.
- **Help in System Testing and Validation:** I was involved actively in the validation process. Utilizing a laptop that had diagnostic software installed, I recorded and monitored the data packets that were being communicated between the locomotive and the station. In a SPAD test, I would cross-check the timestamp of the "Signal Passed at Danger" alarm in the data log with the DMI warning and the following brake application command logged by the On-Board Computer.
- **Documentation Review and Analysis:** I worked through major technical documents such as the Functional Requirement Specification (FRS) and the Safety Requirements Specification (SRS). This gave me detailed insight into the system's logic, including just what conditions trigger each level of braking (service brakes vs. emergency brakes) and the fail-safe measures that operate if there is component failure.

Results and Analysis

The field observations and test data analysis presented strong quantitative evidence of the effectiveness of KAVACH.

- **Brake Application Dynamics:** The performance of the system was impeccable. The Loco KAVACH computer constantly computes a dynamic speed profile as a function of the train's properties and the distance to the subsequent restriction. My examination of the logs revealed that if the train's true speed ever surpassed this computed "braking curve" during travel, the system would instantly trigger service brakes to control the speed, progressing to emergency brakes as needed. This anticipatory braking guarantees the train never exceeds its authority of movement.
- **Location Detection Accuracy:** The tracking function of the system is a highly advanced combination of technology. The RFID tags themselves serve as absolute position indicators. Between the tags, the system employs "dead reckoning," determining the distance travelled through measurements from wheel-mounted tachometers. My research found that this combination produced a location accuracy of ±5 meters, which is crucial for safe operation in highly dense, multi-track environments.
- **Communication Network Strength:** The UHF radio connection was exceptionally strong. Link availability was always greater than 99.9%. This is accomplished using error detection/correction codes

within the data packets and a "store and forward" function within the onboard unit, so that even a temporary loss of communication does not result in a dangerous scenario.

Challenges Encountered and Learnings

The integration of KAVACH into the intricate brownfield setting of Indian Railways is a systems engineering masterclass.

- **Challenge: Integration with Legacy Systems:** The biggest technical challenge was the integration of KAVACH with legacy RI systems. Unlike the direct data output of newer EIs, RI systems needed to be fitted with special "contact proving" relays to monitor the status of the physical signal control relays without any electrical interference that would affect the safety of the original interlocking.

Learning: This taught me the deep significance of interface engineering. I discovered that creating a standardized, electrically isolated, and fail-safe interface is most important when retrofitting contemporary technology on to core legacy infrastructure.

- **Challenge: Radio Network Optimization in Challenging Terrain:** 100% radio coverage is an absolute safety necessity. This was difficult to achieve within areas having deep cuttings, tunnels, and sharp curves that block radio waves.

Learning: I acquired hands-on experience with RF planning. This is more than mere theory and entails extensive on-site surveys with the use of spectrum analyzers to determine dead zones. I learned how to strategically position repeater antennas and how to calculate link budgets to guarantee an adequate signal-to-noise ratio (SNR) to facilitate reliable communication under any circumstances.

- **Challenge: The Human-Machine Interface (HMI) and Trust:** A technologically ideal system is worthless if the human operator does not trust or comprehend it. A major challenge was making the DMI intuitive and unobtrusive for loco pilots used to conventional driving styles.

Learning: I was taught that HMI design is no less important than safety logic. The KAVACH DMI employs a deliberately thought-out combination of visual indicators (color-coded speed bands) and context-aware audio cues. Observing iterative design and pilot feedback sessions emphasized the importance of user-cantered design fundamentals in successful technology uptake.

Conclusion and Future Scope

Summary of Learnings and Skills Acquired

This internship has been a time of vast technical learning, giving a hands-on context to my theoretical knowledge. I have gained a variety of specific skills and an extensive acquaintance with a number of key railway technologies:

- **Technical Skills:** I became practically familiar with the working principles of both Relay Interlocking (RI) and contemporary Electronic Interlocking (EI) systems. I developed the ability to read circuit diagrams, comprehend hardwired safety logic, and value the software-based flexibility of EI. My practical skills further progressed during site maintenance activities, where I developed diagnostic practices for point machines, track circuits, and signalling equipment.
- **Telecommunication Skills:** I gained hands-on experience of the Optical Fiber Cable (OFC) network that is the communication backbone of the railways. This involved seeing techniques such as fusion splicing and the application of an OTDR for fault location.
- **Advanced Systems Awareness:** KAVACH project gave a detailed awareness of Automatic Train Protection (ATP) systems. I got to know about SIL-4 safety levels, the coupling of onboard and trackside units, and how UHF radio communication and RFID technology play the central role in contemporary train control.

Personal and Professional Development

Apart from the technical competence, the internship has played a crucial role in my personal development. Working in the disciplined setup of a government-owned mass transportation organization like Indian Railways has been an enriching experience.

- **Bridging Theory and Practice:** The greatest advancement was seeing the actual application of engineering principles in the field. Observing how theoretical ideas are modified to address practical issues has provided me with a better appreciation of my line of study.
- **Professionalism and Teamwork:** Collaborating with veteran engineers, section supervisors, and technicians provided me with an appreciation of clear communication, cooperation, and compliance with rigid safety measures in a high-risk setting.
- **Problem-Solving:** Witnessing the methodical process of resolving signal failures and integrating new systems such as KAVACH significantly improved my analytical and problem-solving capabilities. This has strengthened my career goals in transportation engineering and safety-critical systems.

Conclusion

Lastly, my internship at the Signal and Telecommunication Department of DRM Office, New Delhi, was an all-embracing and enriching experience. It was a great opportunity to understand the complex mechanisms of the technologies that guarantee the safety and efficiency of Indian Railways. The study of the indigenous KAVACH system in detail was a special highlight, indicating India's capability to develop world-class, safety-critical technology. The internship successfully met all its objectives, providing me with invaluable industry exposure and confirming my passion for the field of railway engineering.

Suggestions and Future Scope

Railway technology is a dynamic field with enormous scope for development in the future.

- **Recommendations for the Organization:** In order to improve future interns' learning experience, a better module could be created that includes fixed sessions for particular legacy and contemporary systems. Affordability of simulation software for interlocking or KAVACH could also provide a secure atmosphere for greater learning.

- **Future Technological Scope:**
- **Improvement of KAVACH:** The development of the system could involve integration with 5G-based FRMCS (Future Railway Mobile Communication System) in terms of increased bandwidth and reduced latency, thus allowing for functionalities such as real-time video observation and enhanced diagnostics. Utilization of AI and Machine Learning over KAVACH data could also facilitate predictive maintenance.
- **Centralized Traffic Control (CTC):** The amalgamation of stand-alone EI and KAVACH-infused stations into a single CTC system is still an area of growth. This will improve traffic flow and enhance line capacity on an overall network basis.
- **Implementation of IoT for Asset Monitoring:** Installation of IoT sensors in signals, point machines, and track circuits can facilitate the transition from schedule to condition-based maintenance, resulting in major cuts in equipment failures and operational expenses.

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